

Many people across African communities, particularly parents, caregivers, pregnant individuals, face difficulties accessing reliable, understandable health information when they need it most. Online health information can be confusing, inaccurate, or difficult for people without medical training to interpret. This may result in misinformation, incorrect self-diagnosis, delayed professional care, or unsafe health decisions. The problem is particularly significant for individuals who lack immediate access to healthcare professionals and need timely guidance about common health concerns.

Our project aims to address this challenge by developing a Small Language Model (SLM) supported by a Retrieval-Augmented Generation (RAG) system that provides concise, evidence-based health information from trusted public health guidelines. The system will explain common health concerns in clear language, identify relevant warning signs, and provide explicit triage guidance indicating whether a situation may be suitable for home management, requires medical consultation, or needs urgent care. The primary beneficiaries are caregivers, parents, pregnant individuals, and people managing chronic or infectious diseases who require fast and trustworthy health information. By improving access to reliable guidance, the project could strengthen health literacy, support informed decision-making, and encourage timely medical intervention.

The project promotes safety, accuracy, accessibility, transparency, empathy, autonomy, and community well-being. However, a significant risk is that users may interpret AI-generated responses as medical diagnoses and rely on them instead of seeking professional care. Other risks include inaccurate, incomplete, or outdated information, cultural differences in healthcare practices, unequal access to appropriate medical services, and increased dependence on AI for health decisions.

To minimise these risks, the system will retrieve information only from authoritative public health sources, provide clear safety disclaimers, avoid diagnosing users, and include explicit escalation guidance for symptoms that may require urgent medical attention. The system is intended to support rather than replace healthcare professionals.

The project's impact also has important limitations and uncertainties. The system cannot account for every individual's medical history, circumstances, or local healthcare context, and its responses depend on the completeness and accuracy of its underlying sources. Users may still misunderstand information or misuse the tool. In addition, the long-term effect of AI-assisted health guidance on public trust and communication with healthcare professionals remains uncertain.

These limitations highlight the need for responsible, transparent, and community-responsive development.

